Samkedemia
EDUCATION

ULTIMATE SOLUTION BOOK:
ALL EXERCISES & PAST EXAMS

Introduction to Methods in Operational Research

KU LEUVEN • BACHELOR OF BUSINESS ADMINISTRATION

VERSION: 4 NOV 2026

LET ME INTRODUCE MYSELF.

About Me

I'll make sure you pass this exam. I know exactly what it takes to get there.



BACKGROUND

- My name is **Jovan Samardzic**, I am 26 years old
- Got my **MSc in Statistics at KU Leuven**, graduated **Magna Cum Laude**
- Recipient of the **Science@Leuven Scholarship**

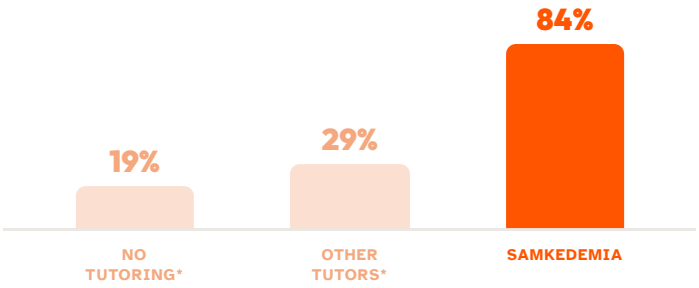
STYLE

- Founder of **Samkedemia**, helping students pass their exams
- Worked with **200+ university students** from all kinds of backgrounds
- **Plain language, not textbook theory** - just what actually makes sense

PROOF THAT IT WORKS.

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* BASED ON INTERNAL RESEARCH CONDUCTED IN 2026.

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Chapter 1 - Indefinite Integrals

Question 1

Show that F is an anti-derivative of f.

(a) $F(x) = \dfrac{x^2}{x+1}, \quad f(x) = \dfrac{x^2+2x}{(x+1)^2}$

(b) $F(x) = x - \ln(1+e^x), \quad f(x) = \dfrac{1}{e^x+1}$

Tip: Verification via Differentiation

To show that $F(x)$ is an anti-derivative of $f(x)$, follow these steps:

1. **Differentiate $F(x)$:** Compute $F'(x)$ using the appropriate rules (Quotient Rule, Chain Rule, etc.).
2. **Simplify:** Algebraically manipulate $F'(x)$ to match the form of $f(x)$.
3. **Conclude:** If $F'(x) = f(x)$, then $F(x)$ is an anti-derivative of $f(x)$.

Solution 1: (a) $F(x) = \dfrac{x^2}{x+1}, \quad f(x) = \dfrac{x^2+2x}{(x+1)^2}$

The function f is defined for all $x \neq -1$. We apply the Quotient Rule:

$$F'(x) = \frac{(2x)(x+1) - (x^2)(1)}{(x+1)^2}$$

$$= \frac{2x^2 + 2x - x^2}{(x+1)^2}$$

$$= \frac{x^2 + 2x}{(x+1)^2}$$

Since $F'(x) = f(x)$, F is an anti-derivative of f.

Answer: $F'(x) = \frac{x^2+2x}{(x+1)^2}$

Solution 1: (b) $F(x) = x - \ln(1+e^x), \quad f(x) = \dfrac{1}{e^x+1}$

Differentiating $F(x)$ term by term:

$$F'(x) = (x)' - (\ln(1+e^x))'$$

$$= 1 - \frac{1}{1+e^x} \cdot (1+e^x)'$$

$$= 1 - \frac{1}{1+e^x} \cdot e^x$$

$$= \frac{1+e^x}{1+e^x} - \frac{e^x}{1+e^x}$$

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Simplifying the expression:

$$F'(x) = \frac{1 + e^x - e^x}{1 + e^x} = \frac{1}{1 + e^x}$$

Since $F'(x) = f(x)$, F is an anti-derivative of f.

Answer: $F'(x) = \frac{1}{e^x+1}$

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Question 2

Show that:

(a) $\displaystyle\int \frac{x^2\,dx}{\sqrt{1+x}} = \frac{2}{15}(3x^2-4x+8)\sqrt{1+x}+C$

(b) $\displaystyle\int \frac{du}{2^u+3} = \frac{u}{3} - \frac{1}{3\ln 2}\ln(2^u+3)+C$

Solution 2: (a) $\displaystyle\int \frac{x^2\,dx}{\sqrt{1+x}} = \frac{2}{15}(3x^2-4x+8)\sqrt{1+x}+C$

Let $F(x) = \frac{2}{15}(3x^2-4x+8)(1+x)^{1/2}$. We apply the Product Rule:

$$\begin{aligned} F'(x) &= \frac{2}{15}\left[(6x-4)\sqrt{1+x} + (3x^2-4x+8)\frac{1}{2\sqrt{1+x}}\right] \\ &= \frac{2}{15}\left[\frac{2(6x-4)(1+x)+(3x^2-4x+8)}{2\sqrt{1+x}}\right] \\ &= \frac{1}{15\sqrt{1+x}}\left[2(6x^2+2x-4)+3x^2-4x+8\right] \\ &= \frac{1}{15\sqrt{1+x}}\left[12x^2+4x-8+3x^2-4x+8\right] \\ &= \frac{15x^2}{15\sqrt{1+x}} = \frac{x^2}{\sqrt{1+x}} \end{aligned}$$

Since $F'(x)$ matches the integrand, the equality is shown.

Answer: $F'(x) = \frac{x^2}{\sqrt{1+x}}$

Solution 2: (b) $\displaystyle\int \frac{du}{2^u+3} = \frac{u}{3} - \frac{1}{3\ln 2}\ln(2^u+3) + C$

Let $F(u) = \frac{u}{3} - \frac{1}{3\ln 2}\ln(2^u+3)$. Differentiating with respect to u:

$$\begin{aligned} F'(u) &= \frac{1}{3} - \frac{1}{3\ln 2}\cdot\frac{1}{2^u+3}\cdot(2^u\ln 2) \\ &= \frac{1}{3} - \frac{2^u\ln 2}{3\ln 2(2^u+3)} \\ &= \frac{1}{3} - \frac{2^u}{3(2^u+3)} \end{aligned}$$

Finding a common denominator:

$$F'(u) = \frac{(2^u+3)-2^u}{3(2^u+3)} = \frac{3}{3(2^u+3)} = \frac{1}{2^u+3}$$

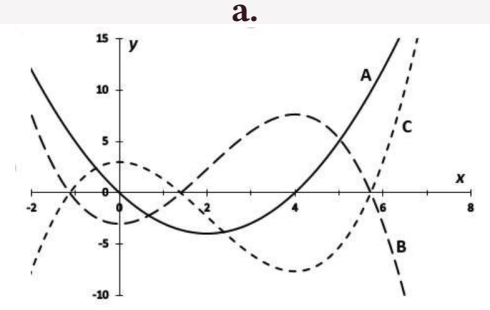
Since $F'(u)$ matches the integrand, the equality is shown.

Answer: $F'(u) = \frac{1}{2^u+3}$

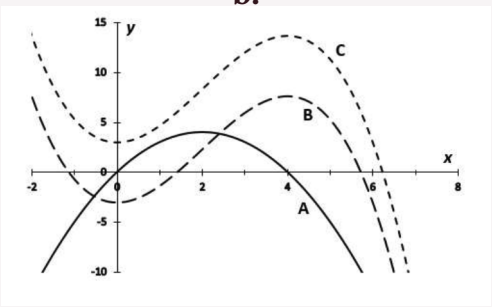
Question 3

In each of the following figures, three graphs are shown. If A is the graph of the function f, which of the other two graphs is the graph of an anti-derivative of f?

a.



b.



Tip: Relationship Between a Function and its Anti-derivative

To identify the graph of an anti-derivative F from the graph of f, follow these steps:

1. **Check the Zeroes:** When the graph of f is at zero (crosses the x-axis), the graph of F must have a local maximum or minimum (a flat turning point).
2. **Check the Sign:** If f is positive (above the x-axis), F must be increasing. If f is negative (below the x-axis), F must be decreasing.
3. **Check the Peaks:** When f has a peak or valley, F changes its curvature (inflection point).

Solution 3: (a) Identifying the anti-derivative for graph A

In figure (a), graph A represents the function f. We look at where graph A crosses the x-axis:

- Graph A is zero at $x \approx 0.5$ and $x = 4$.
- At $x \approx 0.5$, graph C has a local minimum (it turns upward).
- At $x = 4$, graph C has a local maximum (it turns downward).

- Graph B does not have turning points at these specific x-values.

Additionally, between $x \approx 0.5$ and $x = 4$, graph A is positive, and graph C is increasing during this entire interval.

Answer: Graph C

Solution 3: (b) Identifying the anti-derivative for graph A

In figure (b), graph A is a downward-opening parabola representing f.

- Graph A crosses the x-axis at $x = 0$ and $x = 4$.
- Both graph B and graph C have turning points at exactly $x = 0$ (local minimums) and $x = 4$ (local maximums).
- Between $x = 0$ and $x = 4$, graph A is positive, and both B and C are increasing.

Because B and C have the exact same shape and only differ by a vertical shift (a constant C), they both represent valid anti-derivatives of the same function f.

Answer: Both B and C

Question 4

Why is the formula $\int r\,f(x)\,dx = r\int f(x)\,dx$ not correct when $r = 0$?

Tip: Constants in Integration

When working with indefinite integrals and constants, keep these rules in mind:

1. **Integral of Zero:** The integral of zero is an arbitrary constant C, because the derivative of any constant is zero.
2. **Multiplication by Zero:** Multiplying any finished expression by zero results in exactly zero, which removes the constant of integration.

Solution 4

We compare the two sides of the equation when the constant r is zero:

- **Left-Hand Side:** The multiplication happens *inside* the integral before integration:

$$\int 0 \cdot f(x)\,dx = \int 0\,dx = C$$

  The result is an arbitrary constant C, which represents any real number.

- **Right-Hand Side:** The multiplication happens *after* the integration is performed:

$$0 \cdot \int f(x)\,dx = 0 \cdot (F(x) + C) = 0$$

  The result is exactly zero.

Since an arbitrary constant C is not necessarily equal to zero, the two sides are not equal. The formula fails in this specific case because it "loses" the constant of integration on the right side.

Answer: $C \neq 0$

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Question 5

Calculate using integrals of basic functions and basic integration rules:

a. $\int \frac{1}{x^4}\,dx$

b. $\int \sqrt[3]{x}\,dx$

c. $\int (t+1)\sqrt{t}\,dt$

d. $\int (2^x + 3^x)\,dx$

e. $\int \frac{2^x}{3^x}\,dx$

f. $\int \frac{x^3 + x - 1}{\sqrt[3]{x}}\,dx$

g. $\int (\sqrt{x}+1)(x-\sqrt{x})\,dx$

h. $\int \left(x^3 - \frac{7}{x} + e^x\right)\,dx$

i. $\int \frac{\sqrt{t} - t + t^3}{\sqrt[3]{t}}\,dt$

j. $\int (x^e + e^x)\,dx$

k. $\int e^{\sqrt{2}}\,dx$

Tip: Basic Integration Rules

To solve these problems, we apply the basic building blocks of integration.

Power Rule	$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$	$n \neq -1$
Logarithmic	$\int \frac{1}{x}\,dx = \ln\lvert x\rvert + C$	
General Exponential	$\int a^x\,dx = \frac{a^x}{\ln a} + C$	

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Solution 5: (a) $\int \frac{1}{x^4}\,dx$

Rewrite using a negative exponent:

$$\int x^{-4}\,dx$$

Apply the power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$:

$$\frac{x^{-4+1}}{-4+1} + C = \frac{x^{-3}}{-3} + C = -\frac{1}{3x^3} + C$$

Answer: $-\frac{1}{3x^3} + C$

Solution 5: (b) $\int \sqrt[3]{x}\,dx$

Rewrite the radical as a fractional power:

$$\int x^{1/3}\,dx$$

Apply the power rule:

$$\frac{x^{1/3+1}}{1/3+1} + C = \frac{x^{4/3}}{4/3} + C = \frac{3}{4}x^{4/3} + C$$

Convert back to radical form:

$$\frac{3}{4}\sqrt[3]{x^4} + C = \frac{3}{4}x\sqrt[3]{x} + C$$

Answer: $\frac{3}{4}x\sqrt[3]{x} + C$

Tip: Addition, Subtraction, and Scaling

Integration follows linear properties, allowing you to break down complex expressions:

- **Sum and Difference Rule:** You can integrate terms separately:

$$\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$$

- **Constant Multiple Rule:** A constant factor can be moved outside the integral:

$$\int k \cdot f(x)\,dx = k \cdot \int f(x)\,dx$$

Tip: Combining Constants of Integration

When integrating a function with multiple terms, each term technically produces its own constant (C_1, C_2, C_3, etc.). However, since the sum or difference of any constant is still just an unknown constant, we always combine them into a single $+C$ at the very end.

$$\int (x+1)\,dx = \left(\frac{x^2}{2} + C_1\right) + (x + C_2) = \frac{x^2}{2} + x + C$$

Solution 5: (c) $\int (t+1)\sqrt{t}\,dt$

Rewrite $\sqrt{t}$ as $t^{1/2}$ and distribute into the parentheses:

$$\int (t \cdot t^{1/2} + 1 \cdot t^{1/2})\,dt \;=\; \int (t^{3/2} + t^{1/2})\,dt$$

Integrate term by term using the power rule:

$$\frac{t^{3/2+1}}{3/2+1} + \frac{t^{1/2+1}}{1/2+1} + C \;=\; \frac{t^{5/2}}{5/2} + \frac{t^{3/2}}{3/2} + C$$

Simplify the fractions:

$$\frac{2}{5}t^{5/2} + \frac{2}{3}t^{3/2} + C = \frac{2}{5}t^2\sqrt{t} + \frac{2}{3}t\sqrt{t} + C$$

Answer: $\frac{2}{5}t^2\sqrt{t} + \frac{2}{3}t\sqrt{t} + C$

Solution 5: (d) $\int (2^x + 3^x)\,dx$

Apply the exponential rule $\int a^x\,dx = \frac{a^x}{\ln a} + C$ to each term:

$$\int 2^x\,dx + \int 3^x\,dx = \frac{2^x}{\ln 2} + \frac{3^x}{\ln 3} + C$$

Answer: $\frac{2^x}{\ln 2} + \frac{3^x}{\ln 3} + C$

Solution 5: (e) $\int \frac{2^x}{3^x}\,dx$

Rewrite using the property $\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$:

$$\int \left(\frac{2}{3}\right)^x dx$$

Apply the exponential rule with $a = \frac{2}{3}$:

$$\frac{(2/3)^x}{\ln(2/3)} + C = \frac{1}{\ln(2/3)} \cdot \frac{2^x}{3^x} + C$$

Answer: $\frac{1}{\ln(2/3)}\frac{2^x}{3^x} + C$

Solution 5: (f) $\int \frac{x^3+x-1}{\sqrt[3]{x}}\,dx$

Divide each term in the numerator by $x^{1/3}$:

$$\int \left(\frac{x^3}{x^{1/3}}+\frac{x^1}{x^{1/3}}-\frac{1}{x^{1/3}}\right)\,dx \;=\; \int (x^{8/3}+x^{2/3}-x^{-1/3})\,dx$$

Apply the power rule to each term:

$$\frac{x^{11/3}}{11/3}+\frac{x^{5/3}}{5/3}-\frac{x^{2/3}}{2/3}+C \;=\; \frac{3}{11}x^{11/3}+\frac{3}{5}x^{5/3}-\frac{3}{2}x^{2/3}+C$$

Factor out whole powers:

$$\frac{3}{11}x^3\sqrt[3]{x^2}+\frac{3}{5}x\sqrt[3]{x^2}-\frac{3}{2}\sqrt[3]{x^2}+C$$

Answer: $\frac{3}{11}x^3\sqrt[3]{x^2}+\frac{3}{5}x\sqrt[3]{x^2}-\frac{3}{2}\sqrt[3]{x^2}+C$

Solution 5: (g) $\int (\sqrt{x}+1)(x-\sqrt{x})\,dx$

Expand the expression:

$$\int (\sqrt{x}\cdot x-\sqrt{x}\cdot\sqrt{x}+1\cdot x-1\cdot\sqrt{x})\,dx$$

$$\int (x^{3/2}-x+x-x^{1/2})\,dx \;=\; \int (x^{3/2}-x^{1/2})\,dx$$

Integrate using the power rule:

$$\frac{x^{5/2}}{5/2}-\frac{x^{3/2}}{3/2}+C \;=\; \frac{2}{5}x^{5/2}-\frac{2}{3}x^{3/2}+C$$

Answer: $\frac{2}{5}x^2\sqrt{x} - \frac{2}{3}x\sqrt{x} + C$

Tip: Special Cases: Integrals of e^x and Constants

Two fundamental integration rules to remember are:

- **The Exponential Rule:** The integral of e^x is the most straightforward as it remains unchanged:
$$\int e^x \, dx = e^x + C$$
- **The Constant Rule:** The integral of a constant k (any real number) is that constant multiplied by the variable:
$$\int k \, dx = kx + C$$

Solution 5: (h) $\int \left(x^3 - \frac{7}{x} + e^x \right) dx$

Integrate term by term:

$$\int x^3 \, dx - 7 \int \frac{1}{x} \, dx + \int e^x \, dx$$

Apply the power rule, the log rule, and the exponential rule:

$$\frac{x^4}{4} - 7 \ln |x| + e^x + C$$

Answer: $\frac{x^4}{4} - 7 \ln |x| + e^x + C$

Solution 5: (i) $\displaystyle\int \frac{\sqrt{t}-t+t^3}{\sqrt[3]{t}}\,dt$

Rewrite radicals as fractional powers and divide:

$$\int \left(\frac{t^{1/2}}{t^{1/3}} - \frac{t^1}{t^{1/3}} + \frac{t^3}{t^{1/3}}\right) dt \;=\; \int (t^{1/6} - t^{2/3} + t^{8/3})\,dt$$

Integrate each term:

$$\frac{t^{7/6}}{7/6} - \frac{t^{5/3}}{5/3} + \frac{t^{11/3}}{11/3} + C \;=\; \frac{6}{7}t^{7/6} - \frac{3}{5}t^{5/3} + \frac{3}{11}t^{11/3} + C$$

Answer: $\frac{6}{7}t\sqrt[6]{t} - \frac{3}{5}t\sqrt[3]{t^2} + \frac{3}{11}t^3\sqrt[3]{t^2} + C$

Solution 5: (j) $\displaystyle\int (x^e + e^x)\,dx$

We treat the first term using the Power Rule for a constant exponent ($e \approx 2.718$) and the second term using the Exponential Rule for a constant base:

$$\int (x^e + e^x)\,dx = \int x^e\,dx + \int e^x\,dx$$

$$= \frac{x^{e+1}}{e+1} + e^x + C$$

Answer: $\displaystyle\frac{x^{e+1}}{e+1} + e^x + C$

Solution 5: (k) $\int e^{\sqrt{2}}\,dx$

Since $e^{\sqrt{2}}$ is a constant (no x variable in the base or exponent), the integral is simply the constant times x:

$$e^{\sqrt{2}} \cdot \int 1\,dx \;=\; e^{\sqrt{2}}x + C$$

Answer: $e^{\sqrt{2}}x + C$

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Question 6

Compute:

a. $\left(\int \dfrac{x^2+7}{x^2+10}\, dx \right)'$

b. $\int \left(\dfrac{x^2+3x-9}{e^x+\ln x} \right)' dx$

Tip: The "Undo" Trick

For these problems, you don't actually need to do any heavy calculus. You just need to understand that Integration and Derivation are opposites. They "cancel" each other out, like multiplication and division.

Scenario	What's happening?	Result
$(\int f(x)\, dx)'$	Derivative of an Integral	The original function (no changes)
$\int (f(x))'\, dx$	Integral of a Derivative	The original function $+ C$

Solution 6: (a) $\left(\int \dfrac{x^2+7}{x^2+10}\, dx \right)'$

In this case, we are asked to find the derivative (the prime symbol $'$ outside the brackets) of an integral.

Think of this as walking forward three steps and then immediately walking backward three steps. You end up exactly where you started.

The derivative "undoes" the integral entirely. We simply remove both symbols and keep the fraction exactly as it is.

Answer: $\dfrac{x^2+7}{x^2+10}$

Solution 6: (b) $\displaystyle\int \left(\frac{x^2+3x-9}{e^x+\ln x}\right)' dx$

Here, the order is flipped. We are taking the integral of a derivative.

Just like before, the symbols cancel each other out, and we are left with the middle function. However, because the integral symbol is on the outside, we are performing an indefinite calculation.

In business math, whenever the integral is the last thing you do, you must always add the constant $+C$ at the end to represent any fixed values that might have been lost during derivation.

Answer: $\dfrac{x^2+3x-9}{e^x+\ln x}+C$

Booklet 1 - Indefinite Integrals

Chapter 2 - Differential equations

Booklet 2 - Differential equations

Chapter 3 - Definite Integrals

Booklet 3 - Definite Integrals

Chapter 4 - Linear Algebra

Booklet 4 - Linear Algebra

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